AI & ML INTERNSHIP

TASK 2 – EXPLORATORY DATA ANALYSIS (EDA)

Objective	To understand the dataset using statistical methods and visualizations, identify patterns and trends, explore relationships between features, and detect potential anomalies.
Dataset Name	Shopify Sales Dataset (*shopify_sales_dataset_ml_eda.csv*)
Dataset Size	60,000 records and 17 columns
Tools Used	Python, Pandas, NumPy, Matplotlib and Seaborn (Google Colab)

This document contains the executed outputs and visualizations saved in the submitted Google Colab notebook.

1. Dataset Preview

First five rows of the Shopify Sales Dataset.

```
order_id  order_date  customer_id  product_id product_category  \
0         1  2023-04-13        14958        7824      Accessories
1         2  2024-03-11        26825        5557           Sports
2         3  2025-05-10        37450        2225      Electronics
3         4  2023-09-28        20691        7855         Footwear
4         5  2023-04-17        24631        7789           Sports

   product_price  discount_percent  quantity customer_country traffic_source  \
0         143.39                40         1              USA       Paid Ads
1         746.49                15         4              UAE          Email
2         641.06                 5         5           Canada       Paid Ads
3         512.39                 0         3              UAE         Direct
4         415.89                25         3              UAE   Social Media

  payment_method  shipping_cost  rating  is_returned  discounted_price  \
0    Credit Card          21.08     2.0            0             86.03
1         PayPal           5.48     3.7            0            634.52
2      Apple Pay          11.27     3.2            0            609.01
3      Apple Pay          19.22     1.1            0            512.39
4     Debit Card          24.94     4.1            0            311.92

   revenue   profit
0    86.03    64.95
1  2538.08  2532.60
2  3045.05  3033.78
3  1537.17  1517.95
4   935.76   910.82
```

2. Dataset Dimensions

Number of records and columns in the dataset.

```
Total Rows: 60000
Total Columns: 17
```

3. Overall Summary Statistics

Descriptive statistics for numerical columns.

```
order_id   customer_id    product_id  product_price  \
count  60000.000000  60000.000000  60000.000000   60000.000000
mean   30000.500000  30035.075600   4504.738650     403.427068
std    17320.652413  11540.753215   2018.287018     230.045257
min        1.000000  10000.000000   1000.000000       5.010000
25%    15000.750000  20037.750000   2751.750000     203.160000
50%    30000.500000  30079.000000   4503.500000     403.855000
75%    45000.250000  40036.250000   6257.000000     603.135000
max    60000.000000  49999.000000   7999.000000     799.990000

       discount_percent      quantity  shipping_cost        rating  \
count      60000.000000  60000.000000   60000.000000  60000.000000
mean          18.151500      3.009567      13.521758      3.002440
std           12.527663      1.414794       6.627934      1.152265
min            0.000000      1.000000       2.000000      1.000000
25%            5.000000      2.000000       7.760000      2.000000
50%           20.000000      3.000000      13.580000      3.000000
75%           30.000000      4.000000      19.230000      4.000000
max           40.000000      5.000000      25.000000      5.000000

        is_returned  discounted_price       revenue        profit
count  60000.000000      60000.000000  60000.000000  60000.000000
mean       0.148117        330.220212    994.098926    980.577168
std        0.355219        197.114822    806.445785    806.508355
min        0.000000          3.080000      3.080000    -20.490000
25%        0.000000        161.957500    355.620000    341.925000
50%        0.000000        322.560000    760.470000    747.275000
75%        0.000000        482.242500   1462.450000   1449.807500
max        1.000000        799.900000   3999.500000   3989.950000
```

4. Mean Values

Mean of each numerical feature.

```
Mean Values:
order_id            30000.500000
customer_id         30035.075600
product_id           4504.738650
product_price         403.427068
discount_percent       18.151500
quantity                3.009567
shipping_cost          13.521758
rating                  3.002440
is_returned             0.148117
discounted_price      330.220212
revenue               994.098926
profit                980.577168
dtype: float64
```

5. Median Values

Median of each numerical feature.

```
Median Values:
order_id            30000.500
customer_id         30079.000
product_id           4503.500
product_price         403.855
discount_percent       20.000
quantity                3.000
shipping_cost          13.580
rating                  3.000
is_returned             0.000
discounted_price      322.560
revenue               760.470
profit                747.275
dtype: float64
```

6. Standard Deviation

Standard deviation of each numerical feature.

```
Standard Deviation Values:
order_id            17320.652413
customer_id         11540.753215
product_id           2018.287018
product_price         230.045257
discount_percent       12.527663
quantity                1.414794
shipping_cost           6.627934
rating                  1.152265
is_returned             0.355219
discounted_price      197.114822
revenue               806.445785
profit                806.508355
dtype: float64
```

7. Minimum and Maximum Values

Minimum and maximum values for numerical features.

```
Minimum Values:
order_id               1.00
customer_id        10000.00
product_id          1000.00
product_price          5.01
discount_percent       0.00
quantity               1.00
shipping_cost          2.00
rating                 1.00
is_returned            0.00
discounted_price       3.08
revenue                3.08
profit               -20.49
dtype: float64

Maximum Values:
order_id           60000.00
customer_id        49999.00
product_id          7999.00
product_price        799.99
discount_percent      40.00
quantity               5.00
shipping_cost         25.00
rating                 5.00
is_returned            1.00
discounted_price     799.90
revenue             3999.50
profit              3989.95
dtype: float64
```

8. Quartiles

The 25th, 50th and 75th percentile values.

```
25% Quartile:
order_id            15000.7500
customer_id         20037.7500
product_id           2751.7500
product_price         203.1600
discount_percent        5.0000
quantity                2.0000
shipping_cost           7.7600
rating                  2.0000
is_returned             0.0000
discounted_price      161.9575
revenue               355.6200
profit                341.9250
Name: 0.25, dtype: float64

50% Quartile (Median):
order_id            30000.500
customer_id         30079.000
product_id           4503.500
product_price         403.855
discount_percent       20.000
quantity                3.000
shipping_cost          13.580
rating                  3.000
is_returned             0.000
discounted_price      322.560
revenue               760.470
profit                747.275
Name: 0.5, dtype: float64

75% Quartile:
order_id            45000.2500
customer_id         40036.2500
product_id           6257.0000
product_price         603.1350
discount_percent       30.0000
quantity                4.0000
shipping_cost          19.2300
rating                  4.0000
is_returned             0.0000
discounted_price      482.2425
revenue              1462.4500
profit               1449.8075
Name: 0.75, dtype: float64
```

9. Sales Feature Summary

Summary statistics for selected sales-related features.

```
Sales Summary Statistics:
       product_price  discount_percent      quantity  shipping_cost  \
count   60000.000000      60000.000000  60000.000000   60000.000000
mean      403.427068         18.151500      3.009567      13.521758
std       230.045257         12.527663      1.414794       6.627934
min         5.010000          0.000000      1.000000       2.000000
25%       203.160000          5.000000      2.000000       7.760000
50%       403.855000         20.000000      3.000000      13.580000
75%       603.135000         30.000000      4.000000      19.230000
max       799.990000         40.000000      5.000000      25.000000

              rating  discounted_price       revenue        profit
count  60000.000000      60000.000000  60000.000000  60000.000000
mean       3.002440        330.220212    994.098926    980.577168
std        1.152265        197.114822    806.445785    806.508355
min        1.000000          3.080000      3.080000    -20.490000
25%        2.000000        161.957500    355.620000    341.925000
50%        3.000000        322.560000    760.470000    747.275000
75%        4.000000        482.242500   1462.450000   1449.807500
max        5.000000        799.900000   3999.500000   3989.950000
```

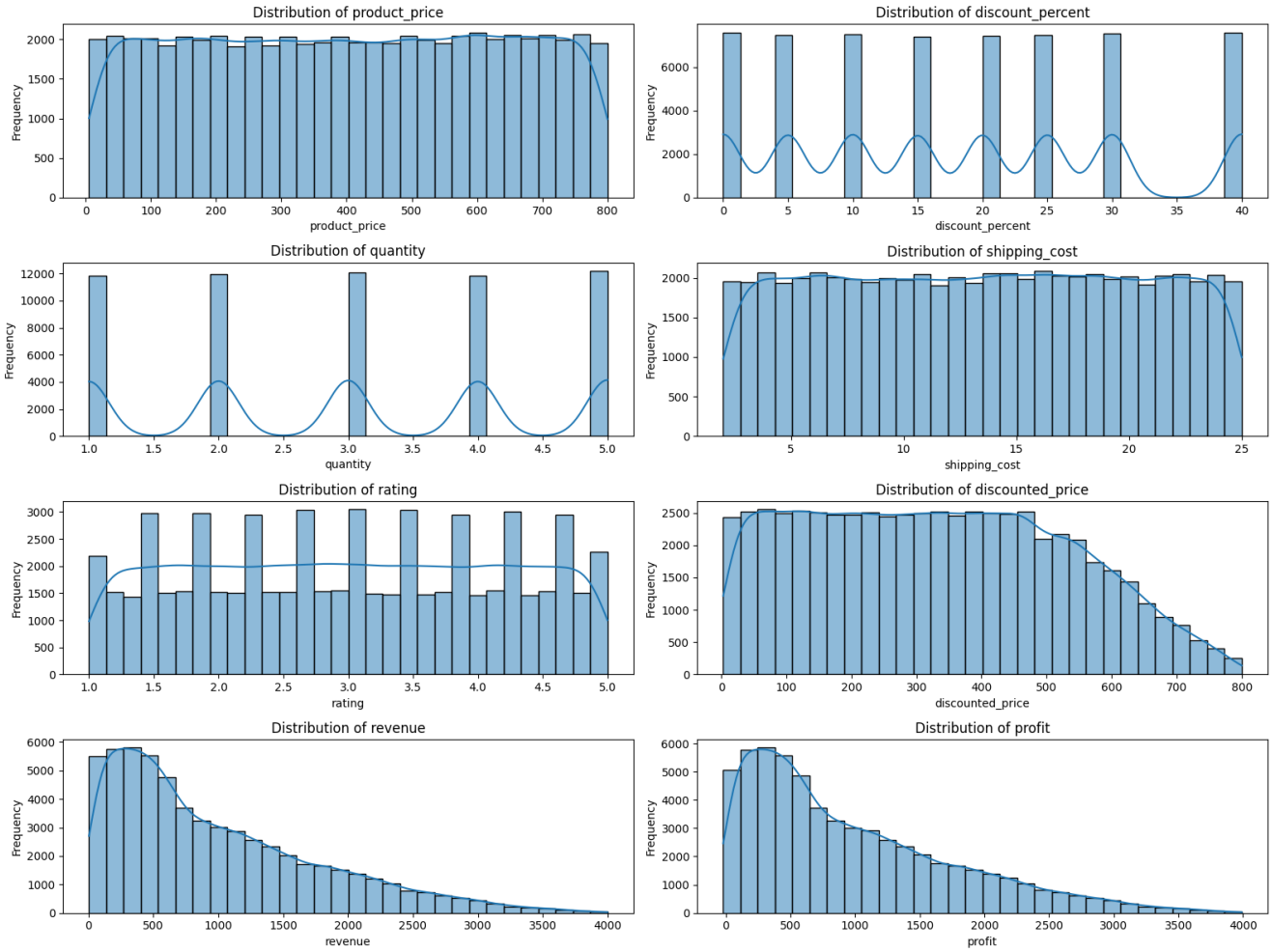
10. Categorical Feature Summary

Count, unique values, most frequent value and frequency.

```
Categorical Data Summary:
       product_category customer_country traffic_source payment_method
count             60000            60000          60000          60000
unique                7                7              5              5
top         Electronics         Australia        Organic      Apple Pay
freq               8624             8714          12075          12116
```

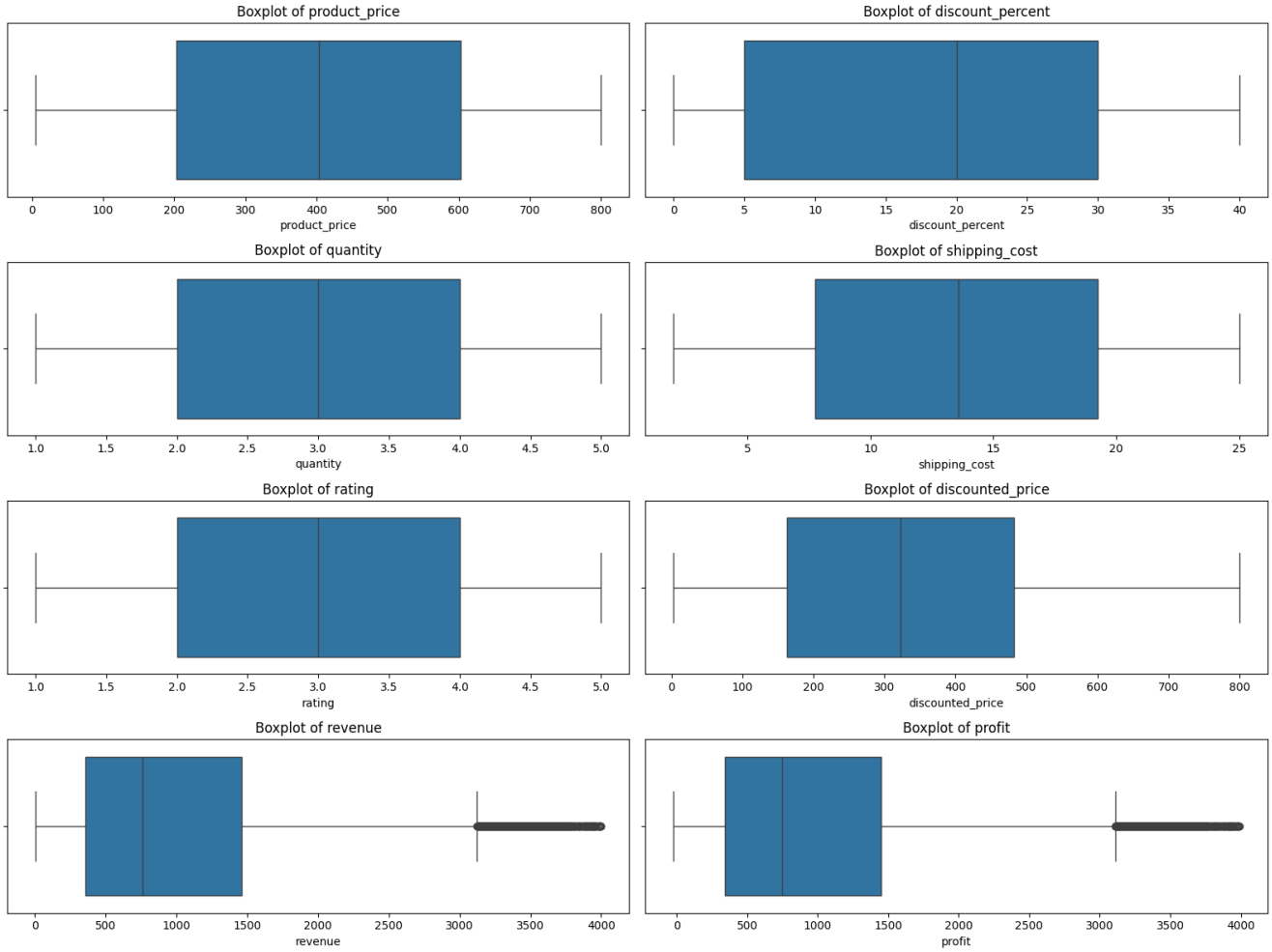
11. Histograms of Numerical Features

Histograms show the distributions of product price, discount, quantity, shipping cost, rating, discounted price, revenue and profit.



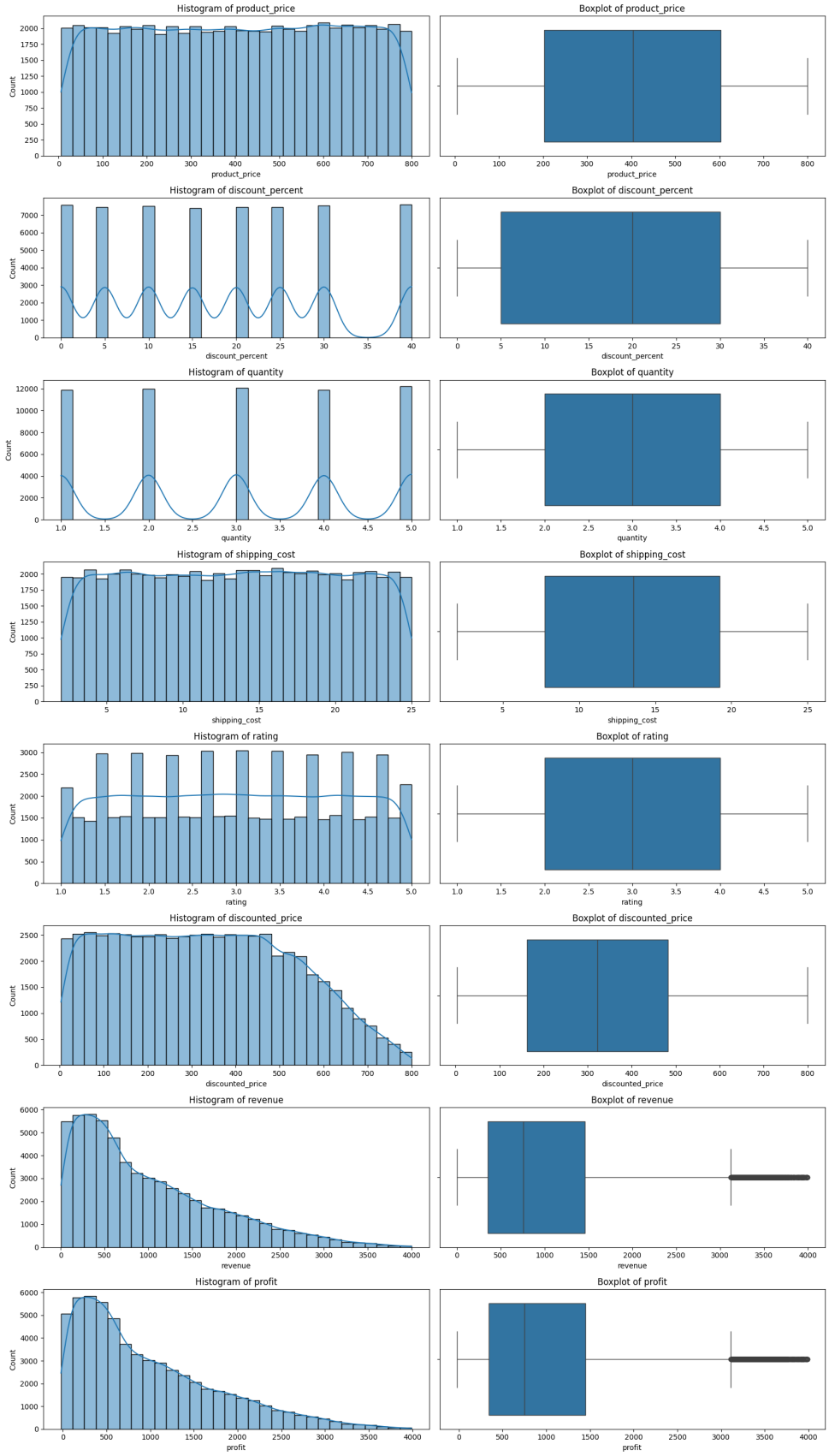
12. Boxplots of Numerical Features

Boxplots summarize the median, spread and potential outliers of the numerical features.



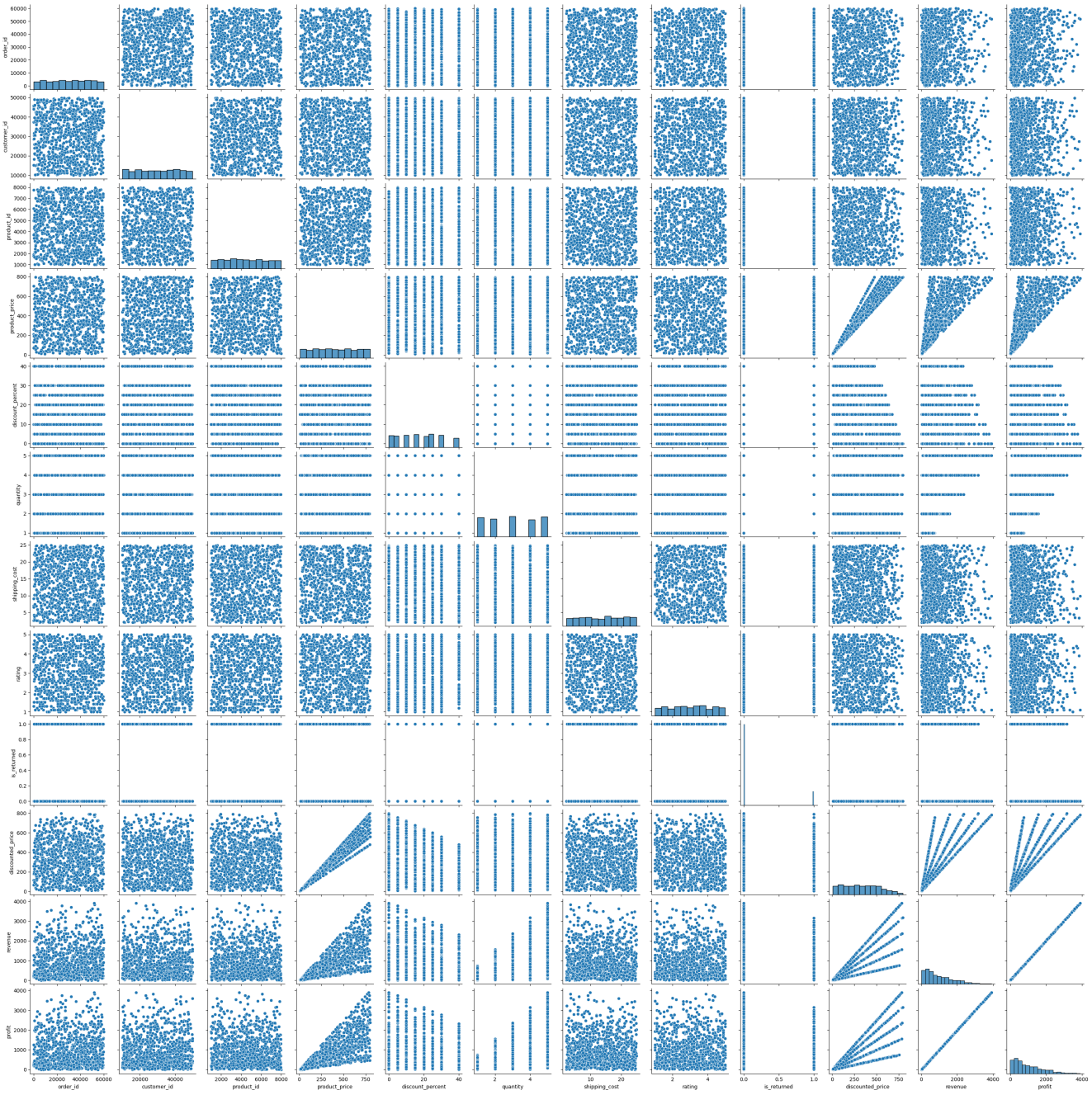
13. Combined Histograms and Boxplots

Side-by-side comparison of the distribution and spread of each numerical feature.



14. Pairplot – Feature Relationships

Pairplot based on a sample of the numerical dataset to visualize pairwise feature relationships.



15. Correlation Matrix Values

Pearson correlation coefficients between numerical features.

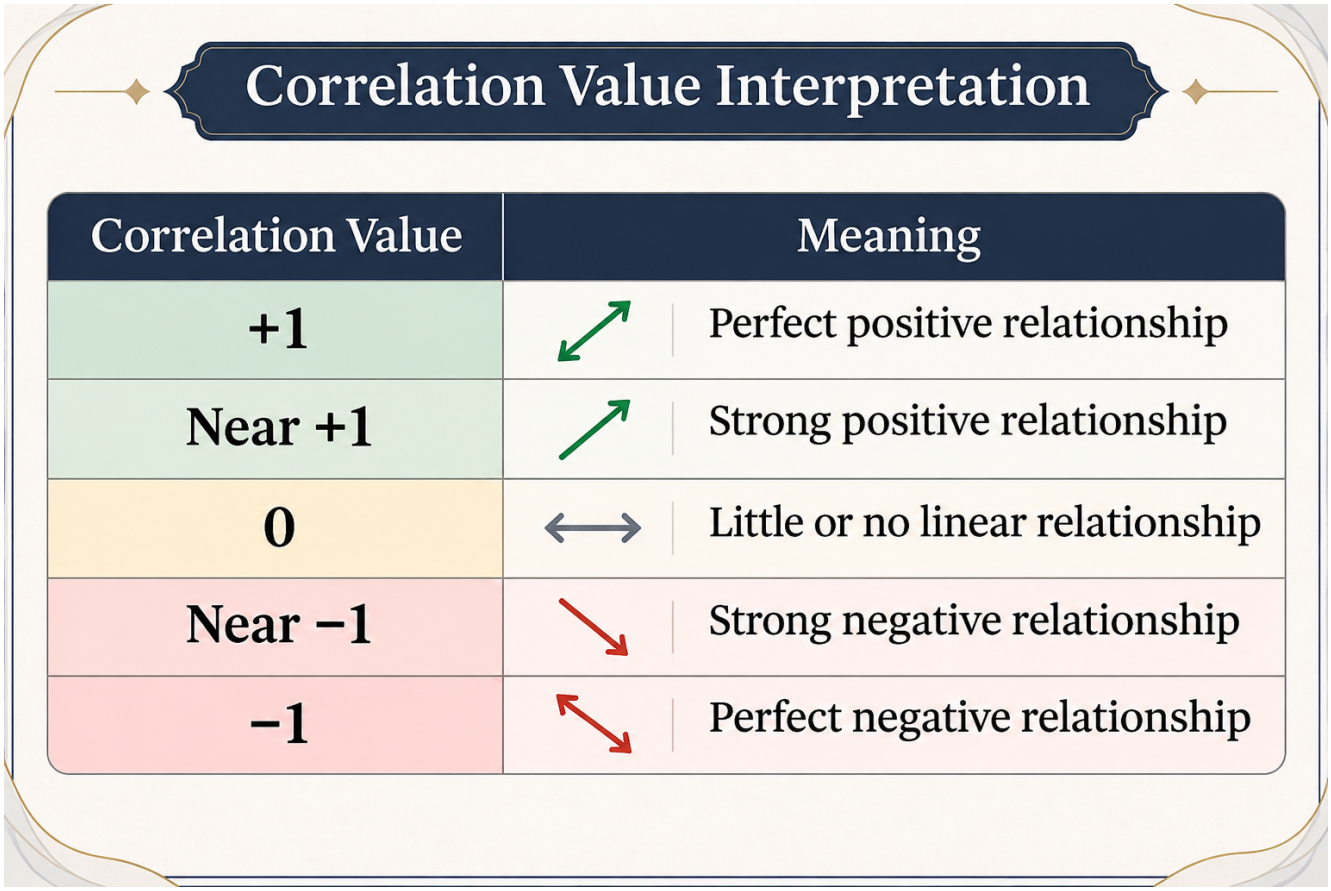
```
order_id  customer_id  product_id  product_price  \
order_id          1.000000    -0.004586   -0.004105      -0.008145
customer_id      -0.004586     1.000000   -0.000534      -0.001576
product_id       -0.004105    -0.000534    1.000000       0.000968
product_price    -0.008145    -0.001576    0.000968       1.000000
discount_percent  0.003079     0.002007   -0.002466      -0.000737
quantity          0.004361     0.001569   -0.003993       0.000847
shipping_cost     0.007272     0.001940    0.005553      -0.005456
rating            0.006561    -0.008395    0.001026      -0.001774
is_returned      -0.005784     0.001642    0.005411      -0.002938
discounted_price -0.008775    -0.002426    0.001685       0.955715
revenue          -0.004984    -0.002784   -0.002394       0.706162
profit           -0.005043    -0.002800   -0.002440       0.706152

                  discount_percent  quantity  shipping_cost    rating  \
order_id                  0.003079  0.004361       0.007272  0.006561
customer_id               0.002007  0.001569       0.001940 -0.008395
product_id               -0.002466 -0.003993       0.005553  0.001026
product_price            -0.000737  0.000847      -0.005456 -0.001774
discount_percent          1.000000 -0.000939      -0.003466 -0.004019
quantity                 -0.000939  1.000000      -0.002925  0.003388
shipping_cost            -0.003466 -0.002925       1.000000 -0.001226
rating                   -0.004019  0.003388      -0.001226  1.000000
is_returned               0.006545 -0.000697      -0.005120 -0.000529
discounted_price         -0.255998  0.001001      -0.004050 -0.001003
revenue                  -0.188741  0.579274      -0.005331 -0.000355
profit                   -0.188698  0.579254      -0.013549 -0.000345

                 is_returned  discounted_price   revenue    profit
order_id           -0.005784         -0.008775 -0.004984 -0.005043
customer_id         0.001642         -0.002426 -0.002784 -0.002800
product_id          0.005411          0.001685 -0.002394 -0.002440
product_price      -0.002938          0.955715  0.706162  0.706152
discount_percent    0.006545         -0.255998 -0.188741 -0.188698
quantity           -0.000697          0.001001  0.579274  0.579254
shipping_cost      -0.005120         -0.004050 -0.005331 -0.013549
rating             -0.000529         -0.001003 -0.000355 -0.000345
is_returned         1.000000         -0.004546 -0.003025 -0.002983
discounted_price   -0.004546          1.000000  0.739114  0.739090
revenue            -0.003025          0.739114  1.000000  0.999966
profit             -0.002983          0.739090  0.999966  1.000000
```

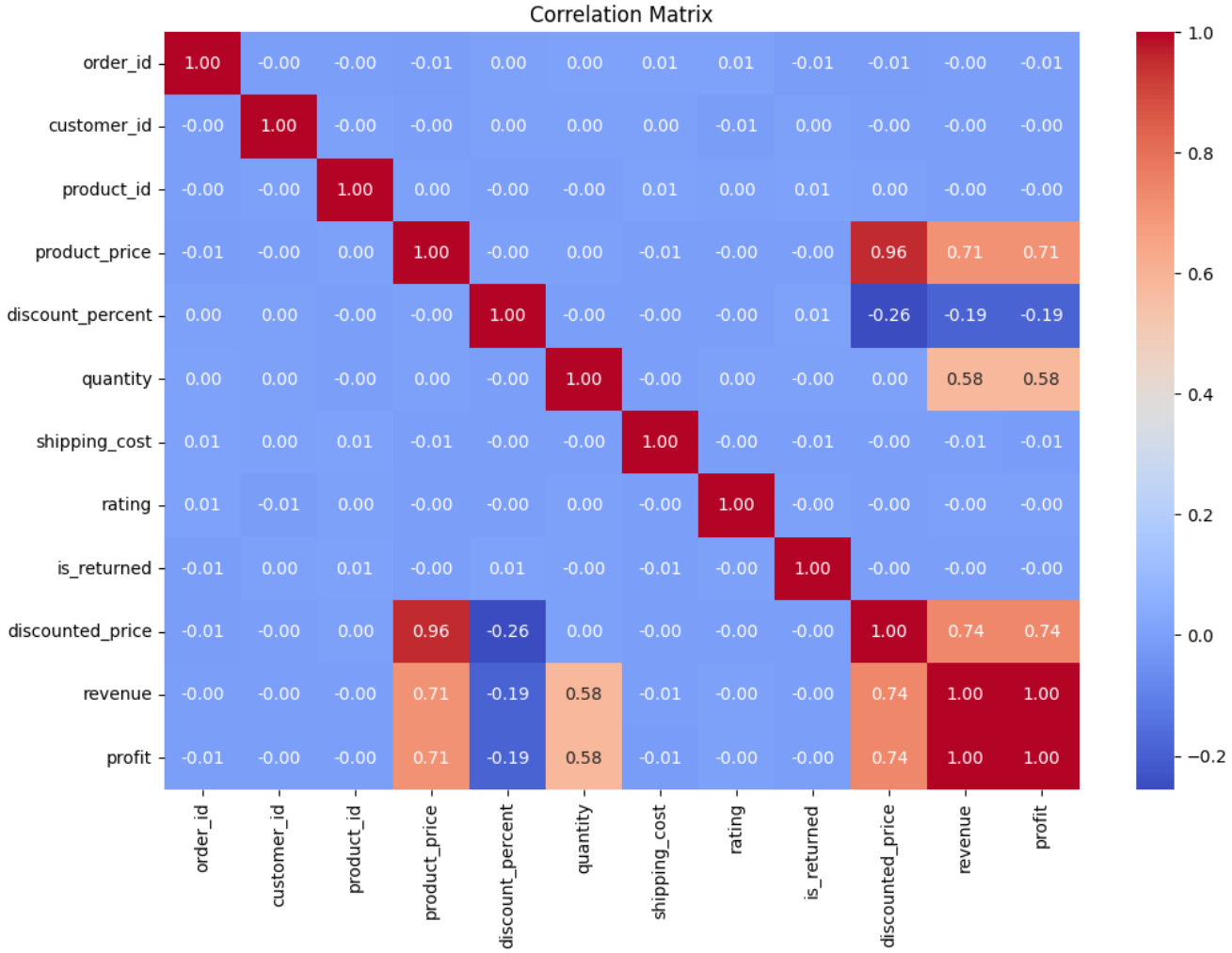
16. Correlation Matrix Output

Correlation matrix included in the notebook.



17. Correlation Heatmap

Heatmap visualizing the direction and strength of linear relationships.



18. Overall Sales Summary

Overall revenue, profit, quantity and categorical summaries.

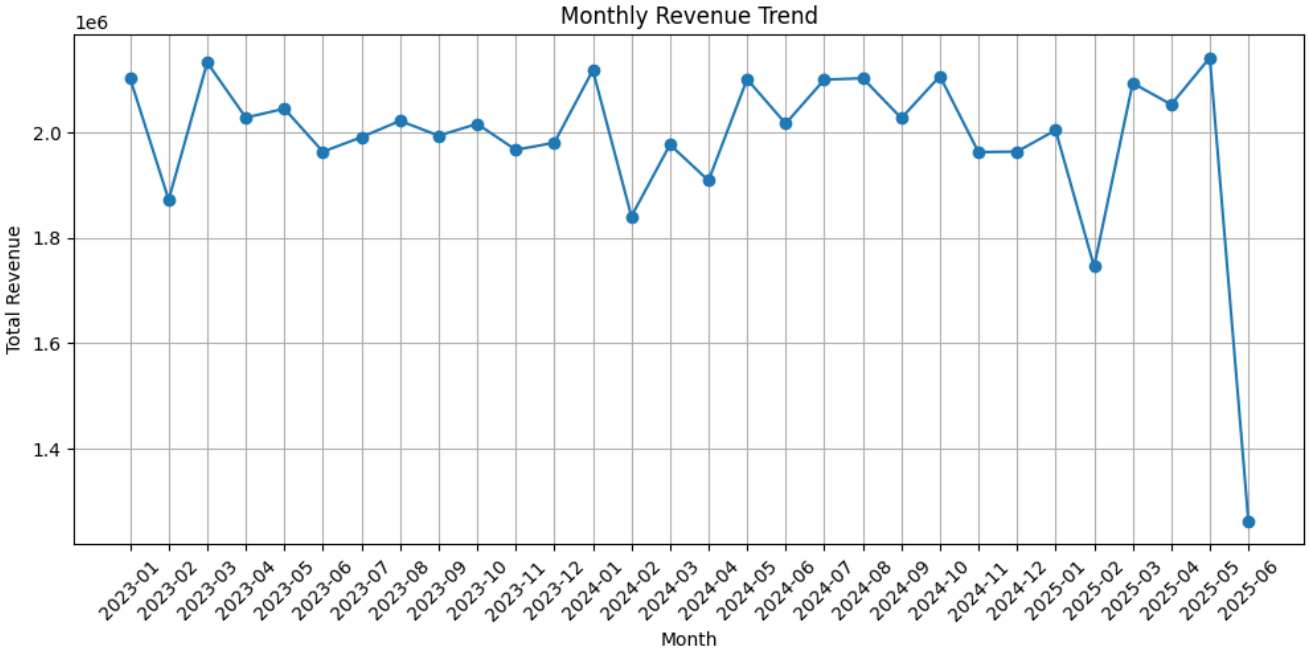
```
Total Revenue: 59645935.58
Total Profit: 58834630.099999994
Average Order Quantity: 3.0095666666666667
Average Product Rating: 3.00244

Product Category Count:
product_category
Electronics    8624
Accessories    8617
Home Decor     8600
Fashion        8594
Beauty         8586
Sports         8522
Footwear       8457
Name: count, dtype: int64

Payment Method Count:
payment_method
Apple Pay           12116
Cash on Delivery    12032
Credit Card         11998
Debit Card          11936
PayPal              11918
Name: count, dtype: int64
```

19. Monthly Revenue Trend

Monthly revenue trend across the available order dates.



20. Revenue by Product Category – Values

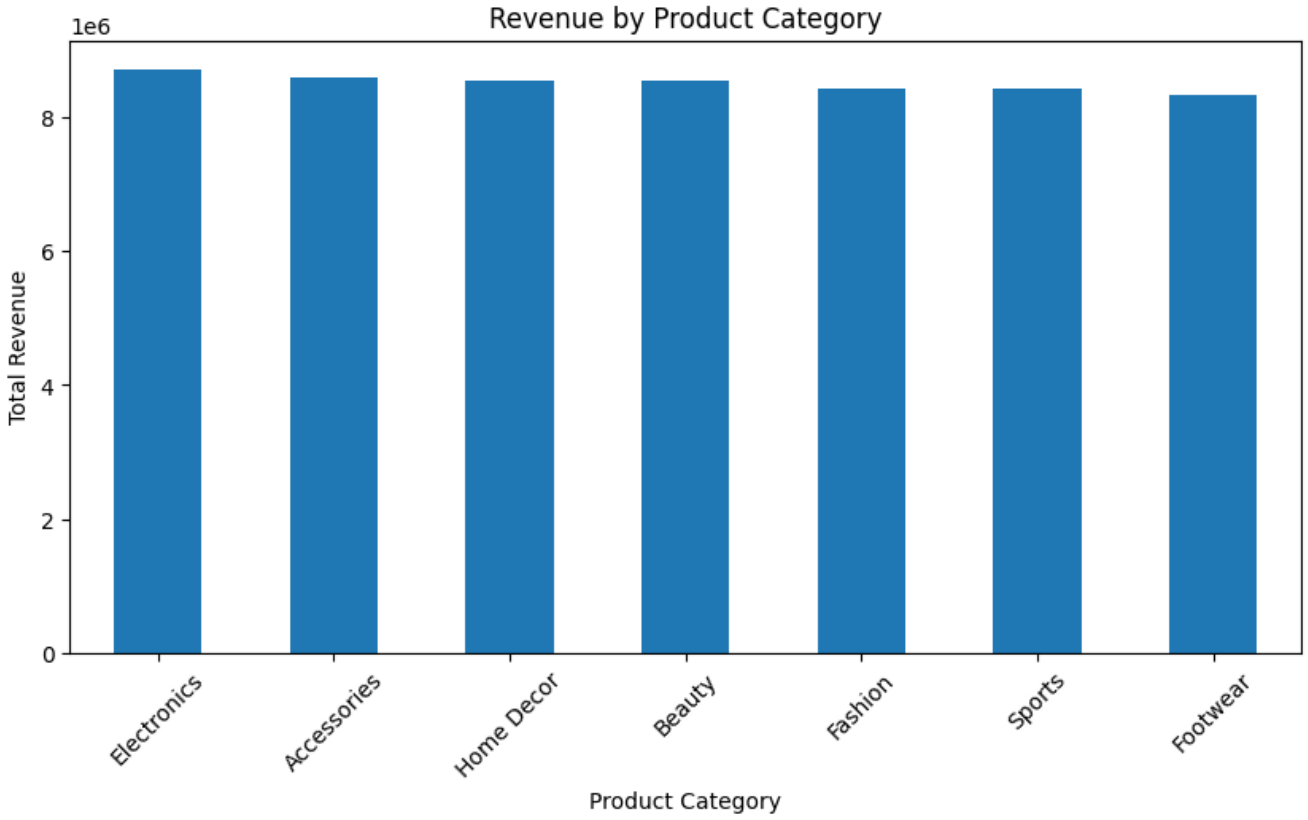
Revenue totals grouped by product category.

```
product_category
Electronics    8717301.49
Accessories    8609171.52
Home Decor     8545840.45
Beauty         8544881.95
Fashion        8445072.67
Sports         8434698.81
Footwear       8348968.69
Name: revenue, dtype: float64

<Figure size 1000x500 with 1 Axes>
```

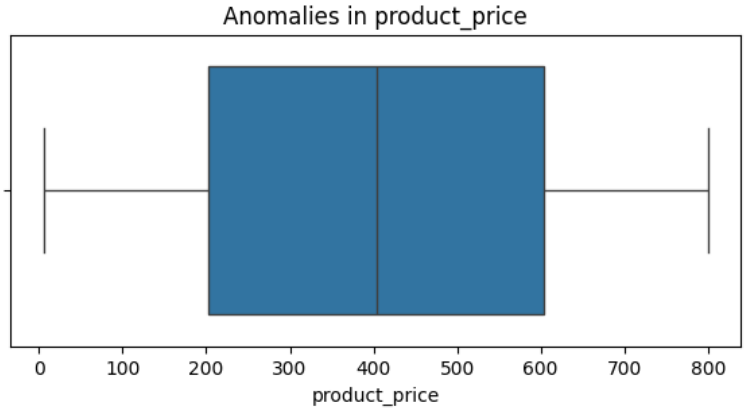
21. Revenue by Product Category – Chart

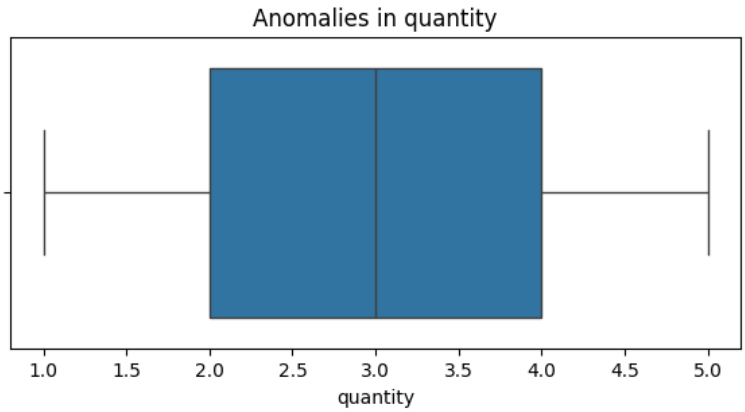
Bar chart comparing revenue across product categories.

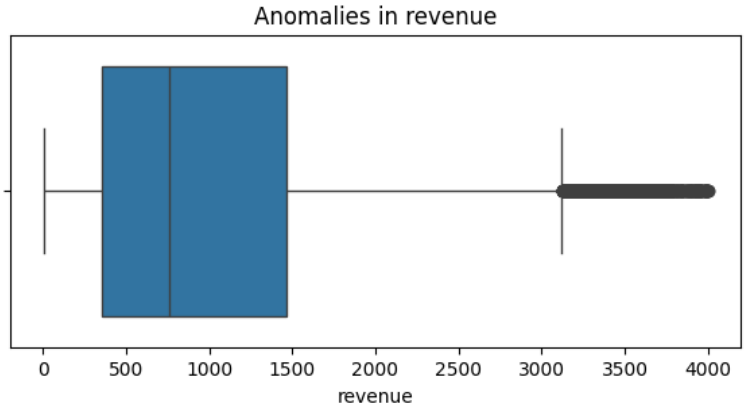


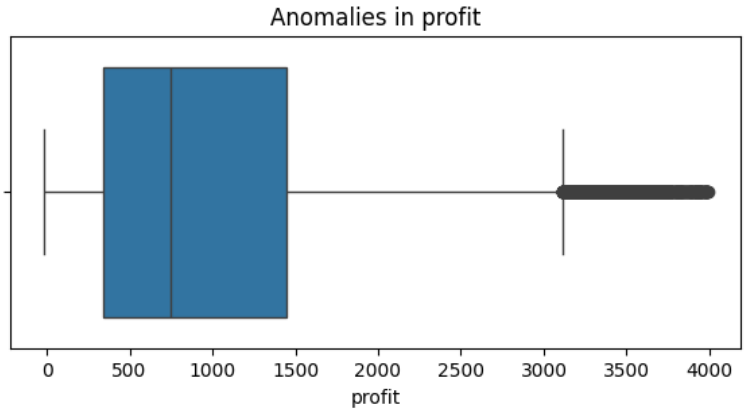
22. Potential Anomalies – Boxplots

Boxplots for selected features used to visually inspect potential anomalies.









23. Potential Outlier Counts

IQR-based counts of potential outliers for selected numerical features.

```
product_price : 0 potential outliers
quantity : 0 potential outliers
revenue : 1014 potential outliers
profit : 1012 potential outliers
```

24. Feature-Level Inferences

Notebook observations about the numerical and categorical features.

```
FEATURE-LEVEL INFERENCES

1. Product Price:
The histogram shows the distribution of product prices.

2. Quantity:
The histogram helps understand the number of products purchased per order.

3. Revenue:
The revenue graph helps identify the variation in sales.

4. Profit:
The profit graph helps understand the distribution of profit.

5. Rating:
The rating graph shows how customers rate products.

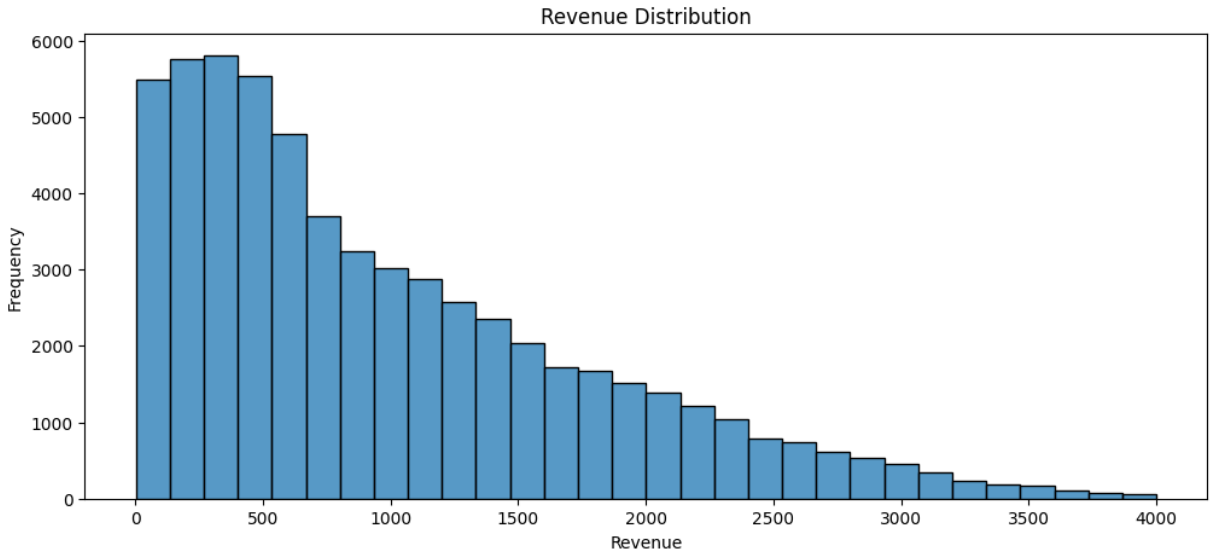
6. Discount:
The discount graph shows the distribution of discounts.

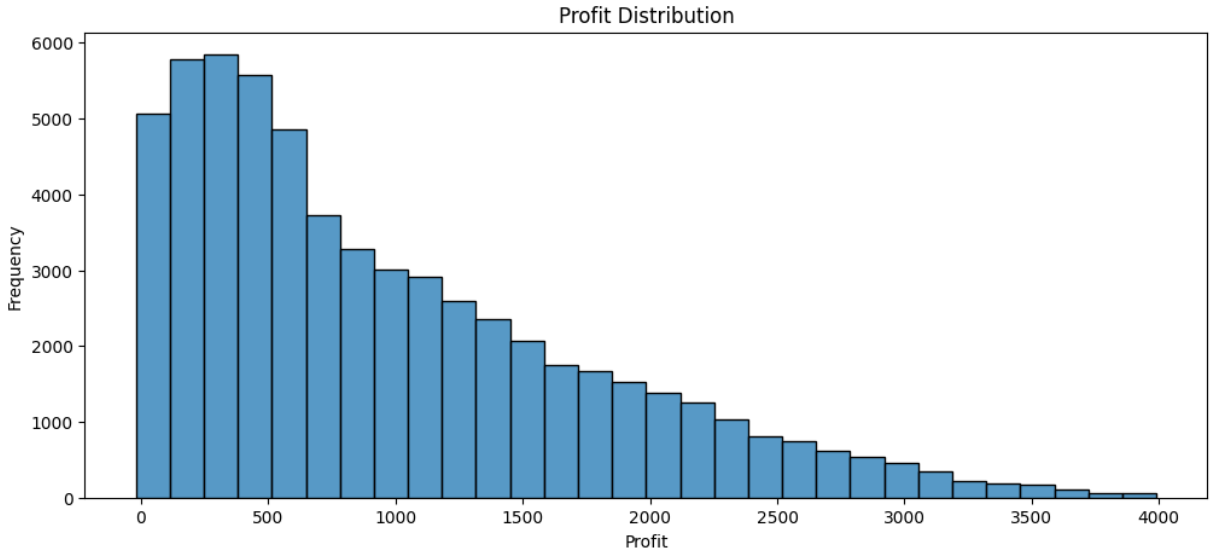
7. Product Category:
The category graph helps identify high-revenue product categories.

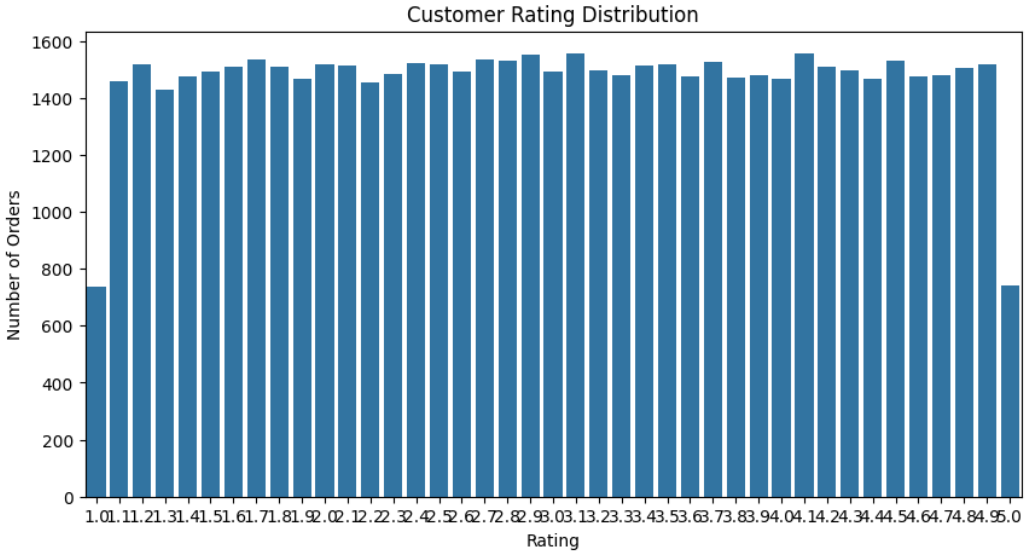
8. Monthly Sales:
The monthly trend graph helps identify changes in revenue over time.
```

25. Revenue, Profit and Rating Distributions

Additional visualizations used to support feature-level interpretation.







26. Final EDA Summary

Final summary generated by the notebook.

```
EXPLORATORY DATA ANALYSIS SUMMARY

Total Records: 60000
Total Features: 18
Total Revenue: 59645935.58
Average Revenue: 994.1
Average Profit: 980.58
Average Rating: 3.0

Most Common Product Category:
Electronics

Most Common Payment Method:
Apple Pay
```